AIMM: Adaptive Information Market Maker

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Abstract

The Adaptive Information Market Maker (AIMM) is a new primitive for prediction markets. It is more capital efficient than traditional automated market makers (AMMs) such as the logarithmic market scoring rule (LMSR), and it predicts more accurately than a parimutuel. Moreover, we show that informed traders need not hold back their information until the last second, as they do in a parimutuel.

1 Introduction

A prediction market elicits information by letting people trade on their opinions. It must solve two conflicting problems:

- capital efficiency for the operator and slippage for the trader;
- the quality of the prediction for the consumer of the information, for instance, an insurance company.

The more capital a market commits, the lower its slippage and the better its prediction.

However, current designs are not optimal:

- AMMs offer good predictions but require a lot of capital;
- a parimutuel needs no upfront liquidity, but its pot does not accurately reflect the true information. Worse, an informed trader holds back until the last second because betting early reveals their information and latecomers can dilute their payout.

Can a market have capital efficiency and information quality that are both good enough?[1]

A prediction market has two components: a pricing rule and a payout rule (Table 1). AIMM resolves the conflict by relaxing both the capital requirement and the accuracy of the information, using a payout rule derived from four axioms (Proposition 1) and the concept of virtual liquidity (Definition 4).

[1] This conflict resembles the uncertainty principle in quantum mechanics: when one is pushed to precision, the other becomes maximally uninformative.

We show that the payout rule and the virtual liquidity together induce an upper and a lower price, and so an upper and a lower probability, around the price quoted by the curve. The length of this probability interval measures the quality of the prediction. For an AMM, the interval reduces to a single point, whereas for a parimutuel it is $[0, 1]$, which is maximally uninformative.

	Pricing rule	Payout rule	Capital requirement
AMM	curve	\$1 per winning token	from the operator
Order book	limit orders	\$1 per winning token	from market makers
Parimutuel	\$1	pot divided by winning tokens	none
AIMM	curve	Proposition 1	Definition 4

Table 1: Pricing and payout rules.

The same structure of upper and lower prices appears in the game-theoretic foundations of probability (Shafer and Vovk 2019) and in several areas of finance (Table 2).

Field	Two prices
corporate finance (Merton 1974)	price and face value
credit pricing (Jarrow and Turnbull 1995)	risky and risk-free prices
incomplete markets (El Karoui and Quenez 1995)	superhedging and subhedging prices

Table 2: Upper and lower prices.

Recall the curve, the pricing rule of an AMM, and its price.

Definition 1 (Curve)**.** *A curve is a smooth, strictly convex, decreasing curve* Γ *in the quadrant* $x, y > 0$ *that approaches the* y*-axis as* $y \to \infty$ *and the* x*-axis as* $x \to \infty$*.*

For example, $xy = k$ is the curve of the constant-product AMM, and $e^{-\frac{x}{b}} + e^{-\frac{y}{b}} = 1$ with $b > 0$ that of the LMSR.

Definition 2 (Reserves and trades)**.** *The market holds reserves* $(x, y) \in \Gamma$ *of YES and NO tokens, starting at* (x_0, y_0)*. A purchase of* n *YES tokens for* m *moves the reserves to*

$$(x + m - n, y + m) \in \Gamma,$$

and similarly for NO.

The operator deposits $\ell \geq 0$ and takes back what the winning tokens leave. Hence, once the market has taken in M and sold $q = (q_Y, q_N)$ tokens, it holds $\ell + M$ of money, and

$$x = x_0 + M - q_Y, \tag{1}$$

$$y = y_0 + M - q_N. \tag{2}$$

Definition 3 (Price)**.** *Let $s \coloneqq -\frac{dy}{dx}$ be the slope of Γ at (x, y). The price of YES is*

$$p \coloneqq \frac{s}{1+s},$$

and the price of NO is $1 - p$.

2 Mechanism design of AIMM

In this section, we give the design of AIMM: its payout rule and its liquidity.

2.1 Payout structure

In this subsection, we derive the payout rule from the following axioms.

1. Solvency: a token is paid only from the money the market holds.
2. No arbitrage: a YES and a NO together are not paid more than $1.
3. Fairness: every winning token is paid the same, whenever it was bought.
4. No withholding: winning tokens are paid as much as Axioms 1 and 2 allow.

Proposition 1 (Payout)**.** *Under Axioms 1 to 4, for $i \in \{Y, N\}$, a winning token of type i is paid*

$$r_i = \min\left(1, \frac{\ell + M}{q_i}\right).$$

Proof. By Axiom 3, every winning YES token is paid the same r_Y. By Axiom 1,

$$r_Y q_Y \leq \ell + M.$$

By Axiom 2,

$$r_Y \leq 1.$$

Hence

$$r_Y \leq \min\left(1, \frac{\ell + M}{q_Y}\right).$$

By Axiom 4, equality holds. Likewise for r_N. □

2.2 Real and virtual liquidity

In this subsection, we define the virtual liquidity and compare it with the real liquidity ℓ.

Definition 4 (Virtual liquidity)**.** *The virtual liquidity L is the most the operator would lose if every winning token were paid $1.*

Proposition 2 characterizes the virtual liquidity.

Proposition 2. *The virtual liquidity is*

$$L = \max(x_0, y_0).$$

Proof. Let the answer be yes. If every winning token is paid \$1, the operator loses $q_Y - M$. By Equation (1),

$$q_Y - M = x_0 - x.$$

Hence

$$\sup(q_Y - M) = \sup_{(x,y)\in\Gamma} (x_0 - x) = x_0.$$

Similarly, if the answer is no, by Equation (2). □

An AMM backs all of L, so its operator must deposit $\ell = L$. AIMM backs only $\ell < L$.

3 Upper and lower probabilities

In this section, we show that AIMM achieves better information quality than a parimutuel, and that an informed trader trades immediately when they have an incentive to do so.

Definition 5 (Upper and lower probabilities)**.** *The upper probability of yes is defined as*

$$\overline{p} \coloneqq \min\left(1, \frac{p}{r_Y}\right),$$

the lower probability of yes as

$$\underline{p} \coloneqq \max\left(0, 1 - \frac{1-p}{r_N}\right),$$

and $[\underline{p}, \overline{p}]$ is the information interval.

Remark. $\overline{p}$ is what a dollar if yes costs, and $\underline{p}$ is 1 minus what a dollar if no costs.

Proposition 3 says that informed traders do not wait.

Proposition 3 (Informed trading)**.** *A risk-neutral trader who believes that the chance of yes is π buys YES exactly when $\pi > \overline{p}$, and NO exactly when $\pi < \underline{p}$.*

Proof. The next YES token costs p and pays r_Y with chance π. Since $\pi \leq 1$,

$$\pi r_Y > p \iff \pi > \min\left(1, \frac{p}{r_Y}\right) = \overline{p}.$$

The next NO token costs $1 - p$ and pays r_N with chance $1 - \pi$. Since $\pi \geq 0$,

$$(1-\pi) r_N > 1 - p \iff \pi < \max\left(0, 1 - \frac{1-p}{r_N}\right) = \underline{p}.$$ □

Remark. In a parimutuel, a dollar on either answer may cost a full dollar, so the information interval is $[0, 1]$, and, by Proposition 3, an informed trader never trades while the market is open (Ottaviani and Sørensen 2009).

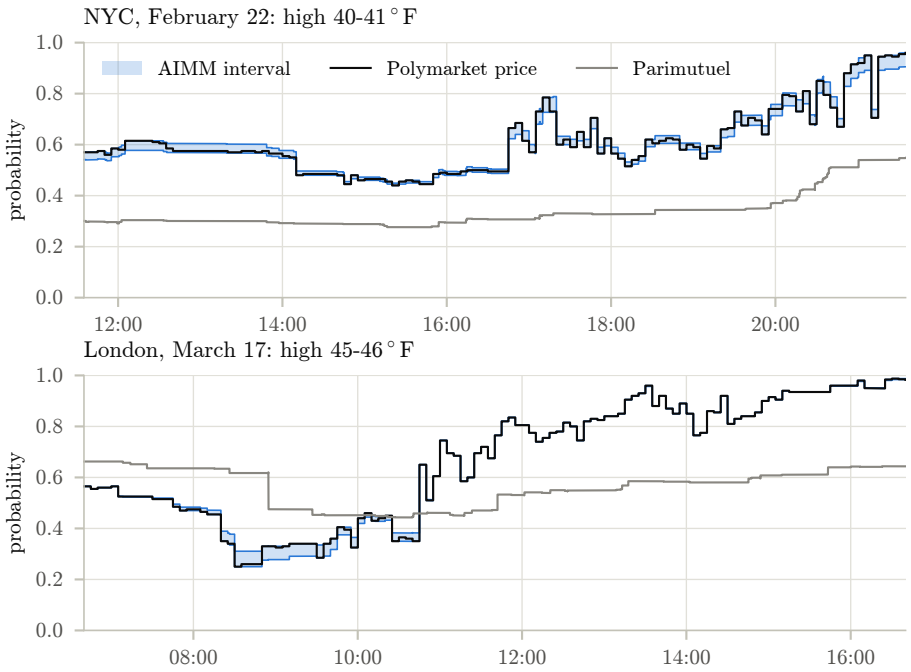


Figure 1: The information interval of AIMM on the curve of the LMSR with real liquidity \$1,000 and $\frac{\ell}{L} = 0.3$, against the price on Polymarket and the share of the pot on yes in a parimutuel that starts with the same \$1,000, in the last ten hours of the highest-temperature markets for NYC on February 22, 2025 and London on March 17, 2025. Every Polymarket trade below \$5,000 also trades on AIMM and on the parimutuel, and a trader who believes Polymarket's price trades as in Proposition 3.

4 Capital efficiency

In this section, we show that AIMM is more capital efficient than an AMM: with the same real liquidity, every purchase moves its price less.

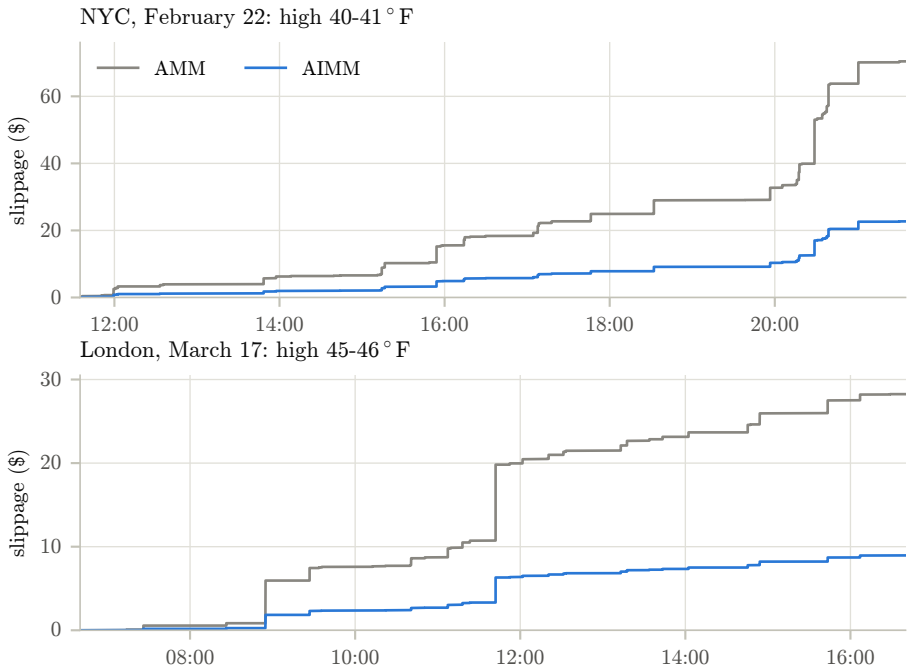


Figure 2: The cumulative slippage, at Polymarket's price, of the trades in Figure 1, on AIMM with real liquidity \$1,000 and $\frac{\ell}{L} = 0.3$, and on an AMM on the curve of the LMSR with the same real liquidity.

Theorem 1 (Capital efficiency)**.** *Let* Γ *have virtual liquidity 1, so that*

$$L\Gamma := \{(Lx, Ly) : (x, y) \in \Gamma\}$$

has virtual liquidity L*, and* $\ell\Gamma$*, defined likewise, has* ℓ*. Let AIMM trade on* $L\Gamma$ *with real liquidity* $\ell < L$*, and an AMM on* $\ell\Gamma$ *with real liquidity* ℓ*. From the same price, every purchase moves the price of AIMM less than that of the AMM.*

Proof. Scaling keeps every slope:

$$\frac{d(Ly)}{d(Lx)} = \frac{d(\ell y)}{d(\ell x)} = \frac{dy}{dx}.$$

Hence, by Definition 3, $L\Gamma$ at (Lx, Ly) and $\ell\Gamma$ at $(\ell x, \ell y)$ quote the price of Γ at (x, y). By Definition 2, a purchase of YES for m takes Ly to

$$Ly + m = L\left(y + \frac{m}{L}\right),$$

and ℓy to

$$\ell y + m = \ell \left(y + \frac{m}{\ell} \right).$$

Since $\ell < L$,

$$y + \frac{m}{L} < y + \frac{m}{\ell}.$$

Since Γ is strictly convex, the price of YES rises with y along it. Therefore the purchase moves the price of AIMM less. Likewise for NO. □

5 Discussion

In this article, we proposed AIMM and showed that it achieves better information quality than a parimutuel and better capital efficiency than an AMM.

In fact, AIMM interpolates between the AMM and the parimutuel (Table 3) and therefore enables the whole design space.

	$\frac{\ell}{L}$	Information
AMM	1	trade at once
Parimutuel	0	wait
AIMM	within $(0, 1)$	trade at once

Table 3: How AIMM interpolates between the AMM and the parimutuel.

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